SeongHyun Kristine Seo

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Research Interests: Probabilistic time-series forecasting · Decision-making under uncertainty · Learning under distribution shift

Highlights

- First-author IEEE Access paper and a registered Korean patent on SHIFT, a self-healing framework for concept drift
- Led three industry–academia projects with POSCO and POSCO DX; four talks at **INFORMS** and **UKC**
- Industry research at **Boeing AI** and **LG Energy Solution**: forecasting, causal discovery, and battery fleet diagnostics

Education

M.S. in Industrial Engineering, Sungkyunkwan University (SKKU) Aug 2025
Thesis: Autonomous online learning with feature store and data store updates: A case study on virtual metrology in steel manufacturing (Advisor: Prof. D.-J. Lim)

B.S. in Data Science and Systems Management Engineering, SKKU Feb 2024

Industry Experience

Diagnostic Algorithm Researcher, LG Energy Solution (CTO Org.) – Seoul, Korea Jul 2026 – Present
- Develop cloud-based anomaly detection and root cause analysis for EV battery field data with GM and Porsche
- Generalize case-by-case diagnostic rules into ML-based methods applicable across product lines
- Determine whether field issues stem from product defects or external factors to support corrective-action decisions

AI/ML Research Intern, Boeing AI (BKETC) – Seoul, Korea Aug 2025 – Mar 2026
- Built an early-warning system to forecast quality risks in aircraft manufacturing and trace their root causes
- Prototyped an end-to-end pipeline from forecasting to automated reporting, refined through iterative deployment

Publication and Patent

Seo, S., & Lim, D.-J.* (2026). SHIFT: Self-Healing Intelligence in Feature store and data store Transitions. *IEEE Access*, 14, 16483–16502.

Lim, D.-J., & **Seo, S.-H.** (2026). System and Method for AI Model Retraining through Feature Store and Data Store Updates. KR Patent No. 10-2941200 (Application No. 10-2025-0049679; filed Apr 2025, registered Mar 2026).

Selected Projects

SHIFT: Self-Healing Model Maintenance under Concept Drift (M.S. Research, Technometrics Lab; POSCO DX)
- Began as a POSCO DX PoC on steel virtual metrology; raised accuracy from 15 to 85 and won approval for production use
- Classified real vs. virtual drift using long/short-window learners and a queue-based decision rule
- Updated features (MDLP + mRMR) and training data (buffer period, weighting); outperformed 11 drift methods

Early Warning Reporting for Aircraft Manufacturing (Boeing AI)
- Built a time-series forecasting pipeline with Prophet to predict delay and defect risk signals
- Quantified directional dependencies with PCMCI and transfer entropy to identify root causes of warning signals
- Designed LLM-based multi-agent workflows that generate and distribute warning-level reports

AI Modeling for BPED Process in LiOH Production (POSCO Holdings)
- Modeled the BPED process for optimization, moving from a Lasso baseline to LSTM-based time-series models
- Detected anomalies in real time by combining domain rule-based statistical thresholds with One-Class SVM
- Led the project as data scientist and PM, delivering a modeling and recommendation platform with process engineers

Predictive Modeling for Productivity Enhancement (POSCO, POSCO DX)

- Built ensemble models (Huber regression, XGBoost, Extra Trees) for five productivity targets in steel manufacturing
- Customized implementations and tuned hyperparameters with plant engineers to fit plant-specific data
- Achieved an accuracy of 84.7, exceeding the project KPI of 80.8

Anomaly Detection with Injection Molding Data (Smart Factory Capstone, Industry–Academia)

- Detected defects with Isolation Forest on Z-score-normalized sensor data for real-time identification
- Built a quality evaluation framework to assess detected defects against production standards
- Automated data labeling for injection molding data in collaboration with an industry partner

Academic Talks and Presentations

Autonomous Online Learning with Feature Store and Data Store Updates: A Case Study on Virtual Metrology in Steel Manufacturing, *Oral*, UKC 2025 – Atlanta, GA — 2025

Automating ML Workflow Orchestration: Strategies for Autonomous Post-Deployment Model Updates, *Oral*, INFORMS Annual Meeting – Seattle, WA — 2024

Steeling Against Time: A Case Study of the Korean Steel Industry, *Poster*, INFORMS Annual Meeting – Seattle, WA — 2024

Is All Well with Your Models? Strategies to Deal with Concept Drift, *Poster*, INFORMS Annual Meeting – Phoenix, AZ — 2023

Honors and Awards

Hyundai Motor Chung Mong-Koo (CMK) Foundation Scholarship, Recipient — 2024 – 2025

Lotte Scholarship Foundation Scholarship, Recipient (5 semesters) — 2021 – 2023

Competitions: 3rd Prize, SKKU Engineering Design Contest (2023); 3rd Prize, SKKU Capstone Design Contest, Team Lead (2023); 2nd Prize, KB Financial Group Solveathon, Team Lead (2022) — 2022 – 2023

Teaching

Teaching Assistant, SKKU – Capstone Design in Systems Management Engineering (2), Undergraduate Research Program, Engineering Research Project, Creative Convergence Design, Fundamentals and Applications of Artificial Intelligence — 2022 – 2025

Skills

Programming: Python, SQL, R
Certifications: Big Data Analysis Engineer, ADsP
Languages: Korean (Native), English (TOEIC 950, OPIc IH), Mandarin (HSK Level 2)